Generative AI in UX Education: Assessing Critical Thinking, Problem Solving, and Design Iteration Speed

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***Abstract*—Recent advancement in generative AI have introduced powerful tools like ChatGPT, Figma AI and UX Pilot that are progressively used in User Experience design processes. These tools can easily generate content, suggest design layouts, and offer interactive feedback, potentially minimizing the need for active cognitive functioning. This raises questions about their impact on design education whether its outcomes prove to be favorable or unfavorable. This study aims to explore how AI technologies have influenced the development of essential abilities in UX education, specifically critical thinking, problem-solving, and design iteration speed using mixed method: qualitative and quantitative. Specifically, by conducting controlled experiments on User Interface and User Experience Design students and gathering feedback through surveys, this study assesses how AI tools have impacted the learning process. The experiment is done by equally dividing the students to design a total of three different UX design tasks, first half with the assistance of AI and the other without. The design outputs were evaluated under a scoring system based on UX principles and usability heuristics. The results revealed that AI primarily enhanced the iteration speed and design process but did not necessarily have impact on the usability of the final results or improvement of the students' problem-solving and critical thinking. Hence, while AI remains crucial for efficiency in the UX education, critical thinking and ethical considerations should be taught alongside in order for students still learn to think on their own accord and to avoid relying solely on the generated results.**

***Index Terms*—Generative AI, UX/UI Education, Problem Solving, Critical Thinking, Design Process, Human-AI Collaboration.**

I. Introduction

Generative Artificial Intelligence (AI) is rapidly transforming the way digital products are form and created, particularly in the field of User Experience design. As tools like ChatGPT, UX Pilot, and AI-integrated design platforms such as Figma AI become commonplace, educators and students alike are compelled to integrate them into UX workflows in order to achieve efficiency and accelerate the design processes. This research investigates a timely question: how does the adoption of generative AI in UX education affect students' critical thinking, problem-solving ability, and design speed? This study mainly argues that generative AI can increase the efficiency of design production and ideation, yet it presents challenges for nurturing independent thoughts and creativity; skills that are vital for UX design. To develop UX course effectively, it is essential to understand how these technologies contribute or interfere with knowledge building.

As there has been a growing interest in using AI tools in education, several empirical studies have been made in the recent years by assessing the impact of AI on education, especially in cognitive skills of students. Although studies like Feng et al. [1] and Muehlhaus et al. [2] have demonstrated the capability of AI in assisting design projects, the specific educational areas for how generative AI could have an impact on the cognitive development of design students remains a underexplored area. Therefore, our study mainly concentrate on how AI affects on UI/UX students' creative and critical thinking which is on the educational site. To explore this issue, this study uses a mixed-methods approach that includes facilitated experiments with hands-on design tasks along with pre and post evaluation surveys to aim to address the two leading questions:

- RQ1: Is there a significant difference in usability of AI-assisted UI designs compared to non-AI assisted designs?
- RQ2: To what extend does AI support tools affect the problem solving, critical thinking, creativity and efficiency of the design process?

II. Literature Review

The development of Generative AI in the recent years have shown revolutionary impacts on UX education. One 2025 survey stated that majority of IT professionals primarily used AI tools in performing their jobs [3]. Despite the increased adoption of AI into UX Education, extending the possibility of ideation, prototyping and research assistance, the integration also raised ethical concerns regarding bias, ownership, and the balance between human creativity and automation. The review explored the role of generative AI in UX education by using the following core themes:

A. Integration of AI in UX Education

The use of AI has grown rapidly nowadays and so in UX education. Despite having some disadvantages of using AI in UX/UI, it continues to be implemented for its benefits of enhancing creativity, ideation and design workflow in various industries in terms of UX designs. According to previous research, AI has been integrated in UX research and practical education [4]–[6]. These studies show that incorporating AI tools in UX design courses can encourage students to develop strategic and innovative ideas [4], [5]. Additionally, applying AI tools in UX research can accelerate with repetitive tasks; data transcribing, surveys, idea generation and analysis [6].

B. Benefits of AI in UX Education

In numerous studies, AI impacts positively on learning and design outcomes in UX education. The integration of AI allows the capacity of iteration process [7], expedited prototyping and iterative testing processes [8], an extensive design possibilities [9]. A study [10] uncovered that using generative AI in education was positively correlated with both digital design thinking skills and academic creativity, especially among students with high technological literacy.

Incorporating AI tools like ChatGPT, Midijourney, Runway, and DALL-E 2 supports design-based learning, development of persona, and ideation, while enhancing the creative self-efficacy and reflective thinking despite no measurable effect on the design thinking mindset [11], [12]. AI allows efficient generation of design which significantly reduces the time taken [13].

Further studies highlight that using AI in UX education can bring educational and psychological benefits; enhancing cognitive creativity with self-efficacy and reductions in anxiety [14]. Moreover, it can improve students' positive attitudes and high interest to AI integration in the UX/UI design processes [15].

C. Limitations and Concerns Regarding AI in UX Education

In contrast to all the advantages, the adoption of AI in UX education came with its challenges. Increased reliance on the AI tools could cause to limitation of the thinking capability of human, which progressively lead to lack of interest and engagement [16]. A study conducted in 2024 [17] revealed that 83% of respondents expressed their concerns towards the excessive reliance on AI might weaken their ability to think independently and make responsible choices. Moreover, researchers expressed that AI-generated contents could lack in human-centered perceptive [18] in the characteristics of UX, which reinforce stereotypes instead of empathy-building with users [19].

III. Methodology

A. Research Design and Procedure

The purpose of the study is to examine the impact of generative AI in UX Education, particularly in assessing critical thinking, problem-solving and design iteration speed during design process. Therefore, the mixed-methods [20] experimental design was employed, to quantitatively measure performance differences with accurate, numerical data, and to qualitatively understand and explain the detailed descriptions of people's experiences.

The study was conducted in three phases: (1) pre-experiment survey, (2) controlled experiment, and (3) post-experiment survey. Pre-experiment surveys captured the demographic information of the participants, prior experience with the usage of AI, creativity and the attitudes toward ethical concerns.

To perform the controlled experiment, a few students who answered the previous pre-experimental survey were recruited and randomly assigned into two groups of students: one using generative AI tools such as ChatGPT and UX pilot, with another one utilizing traditional design methods in the UX/UI design workflow. This aimed to analyze and assess the differences in the outcome quality in terms of creativity, problem solving, critical thinking and the efficiency of design iteration. Each participants were instructed to complete three Figma UI design tasks with increasing complexity levels.

Post-experimental survey collected the experience of the students who performed the tasks in the controlled experiment. They were asked Likert scale and open-ended questions to capture their opinion and experience during the experimental phase.

B. Participants and Sampling

The target population of the research is the UX/UI undergraduate students who were studying Software Engineering at Mae Fah Luang University, Thailand. The research was approved by Ethics Committee on Human Research department of Mae Fah Luang University (Protocol No. EC 25107-13).

First, a large number of students were recruited for the initial data gathering of pre-experimental phase. As the total target population is N=170, the appropriate sample size (n) for pre-experimental survey was calculated using Yamane's simplified formula [21], thus, resulting in 119 students.

Then, among the 119 students who participated in the initial gathering of pre-experimental data, 12 students were selected by purposive sampling [22] for the controlled experiment. From the pool of 119 pre-experimental survey takers, 12 students were selected and examined for the purpose of capturing rich and qualitative feedback, while remaining the in-depth experimental analysis under the logistical constraints of conducting in-depth experimental sessions. Participants are

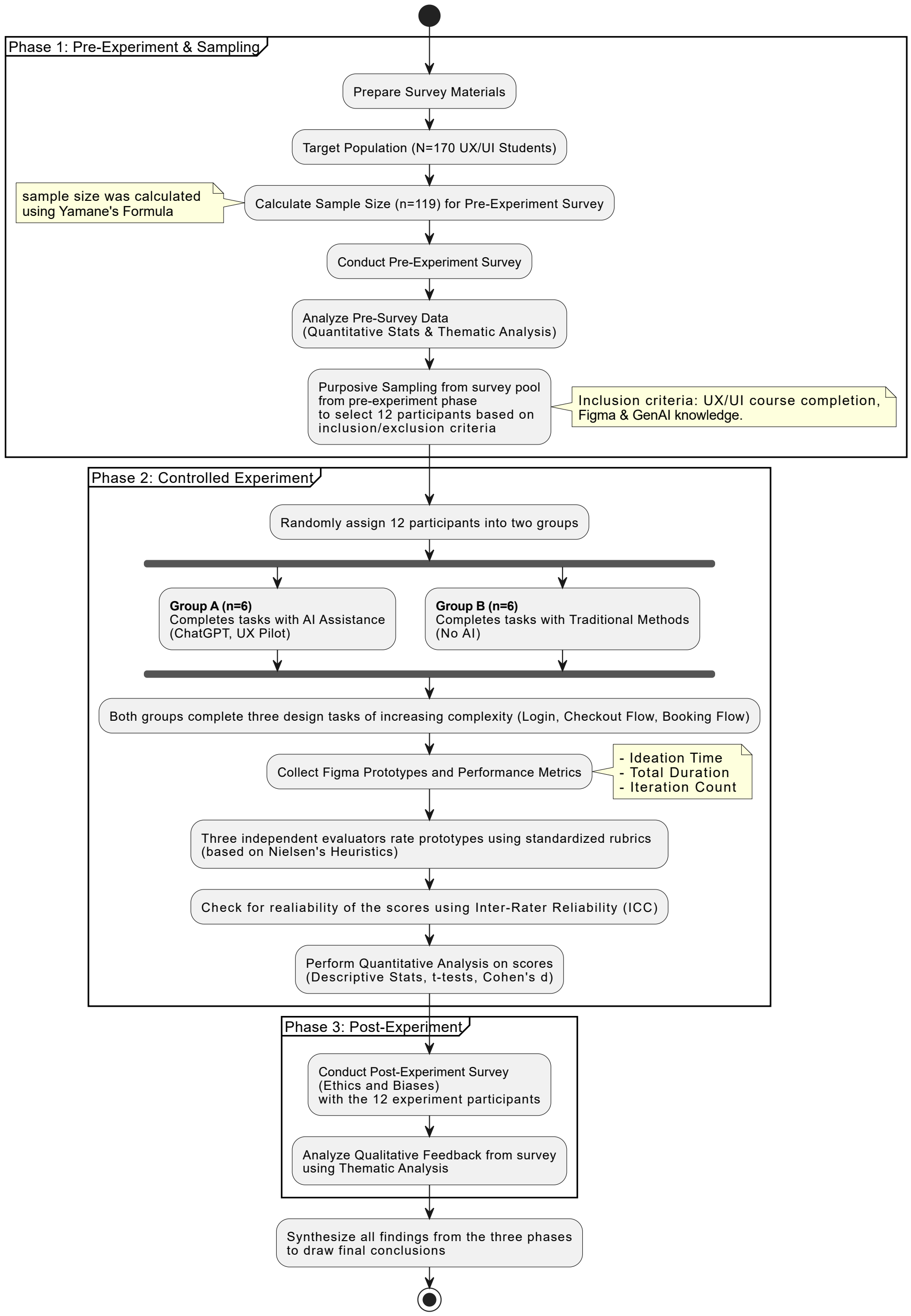


Fig. 1: Research Methodology Flowchart

eligible for inclusion if they have already completed at least one introductory UX/UI design course, and if they have a good knowledge of using UI tools such as Figma and Generative AI tools. Additional inclusion consists of students who can commit fully to the study during the research period. Voluntary students consents were also required. The exclusion criteria included students with with no prior exposure to UX/UI design fundamentals, and students who were unwilling or unable to use AI-based design tools such as Figma. Students who were unable to provide informed consents were also excluded. Participants were recruited through various online channels such as university E-mails, Google Classroom, in addition through the posters displayed on the notice boards around the campus.

C. Data Collection and Analysis

Pre-experiment survey was conducted to gather then general perception of the UX/UI students on the usage of AI in UX/UI education. The data collected from all 119 participants during the pre-experiment survey phase were cleaned. Quantitative data was prepared for the statistical analysis, while qualitative data from open-ended questions were prepared for thematic analysis [23].

Regarding the controlled experiment, all 12 selected participants were divided randomly into two groups; one with AI and another without AI. ChatGPT and UX Pilot were used as AI tools in the study. The participants were then instructed to design the three task scenarios independently as follows:

1) Task 1 (Low complexity) - included one page login of an online bookstore application ”bookworm”
2) Task 2 (Medium complexity) - included three pages checkout flow of an e-commerce application called ”Be loved”, with each screen was instructed as follows:
   a) Shopping cart Screen
   b) Shipping and Payment Screen
   c) Order Confirmation Screen
3) Task 3 (High complexity) - included three pages of a public transportation booking app “iBus”, with each screen was instructed as follows:
   a) Home/Search Page
   b) Route Results Page
   c) Detailed Route Page

Then, the prototypes they developed were rated by the three non-participant evaluators, by using different evaluation rubrics based on Nielsen 10 usability heuristics [24] as shown in Table I. Ratings were checked for reliability prior to analysis, using inter-rater reliability method called Inter-Class Correlation ICC(2,1) and ICC(2,k) [25], and then later examined quantitatively by descriptive statistics such as the mean (M) and standard deviation(SD), and statistical methods such as t-test [26] for each group and task.

TABLE I: Evaluation Rubrics and Rating Scores

Evaluation Rubric	Sub-Criteria	Max Score
Usability (25)	Navigation clarity	5
	Consistency	5
	Accessibility	5
	Aesthetic & minimalist design	5
	Recognition rather than recall	5
Problem Solving (15)	Task completeness	5
	Efficiency	5
	Logical flow	5
Critical Thinking (10)	Problem framing	5
	Justification & reasoning	5
Creativity (15)	Novelty	5
	Appropriateness	5
	Integration	5

Regarding the post-experimental survey, qualitative data from the open-ended questions were analyzed using thematic analysis [23] to address the recurring themes in the participants’ experience and perceptions.

IV. Results

The outcomes from the mixed-method study are presented in this section, starting with the pre-experimental results followed by the controlled experiment and concludes with the post-experimental survey.

A. Pre-experiment Survey

1) Demographics and AI Usage Frequency: The majority of participants were from Year 3 (40.3%) and Year 4 (40.3%). The remaining participants (19.3%) were Year 2 students. When asked the frequency of the usage of generative AI tools, about 42% of participants use AI “Often” and 28.6% as “Always”, while 17.6% and 10.9% stated as “Sometimes” and “Rarely” respectively. Most respondents (72.3%) used AI tools for ideation and brainstorming, followed by the usage of AI for researching users’ needs (58%) and for generating UX/UI elements (54.6%). The remaining purpose of utilizing AI tools was to write UX copy and texts with 35.3%.

2) Perceived Impact on Performance and Efficiency: Majority of respondents acknowledged the improvement of performance and efficiency of design phase using AI tools, with nearly half of them agreed on its user-friendliness as well.

The larger proportion of survey takers (77.3%) agreed or strongly agreed that AI tools speed up the overall design iteration speed, while minority (18.5%) remained neutral and a few (4.2%) disagreed. Specifically on initial prototypes, most of them (69.7%) agreed that AI helps create initial prototypes faster, while minority (23.5%) remained neutral and a few (5.9%) disagreed. Regarding the user-friendliness of the AI generated results, nearly half of the participants (45.4%) stated AI designs are as user-friendly as personally curated designs, even though neutral made up to 37.8% of the samples, and the remaining (16.8%) disagreed.

3) Views on Critical Thinking and Creativity: Majority of the participants believed it is necessary to double-check AI-generated results due to their skepticism towards reliability, despite AI assistance in creative AI design solutions.

Quantitative Analysis: Nearly half of participants (47.9%) agreed or strongly agreed that using AI tools made them likely to critically evaluate the ideas, while 42% remained neutral, and the remaining (10.1%) stated their disagreement. Additionally, about half of participants (49.6%) were neutral regarding the accuracy and reliability of AI-generated works, while 31.1% trusted and 19.4% of them distrusted AI generated designs. Therefore, 83.2% felt the need to double-check AI-generated contents.

In terms of creativity, most (58.9%) agreed that generative AI tools helped with more creative design solutions. In the meantime, 27.7% remained unbiased, and 13.5% was against the claim.

Qualitative Analysis: When asked the situations when participants trusted or mistrusted the AI generated designs, different opinions are received. Most participants trusted AI for initial ideations, and basic information, while many other participants saw AI as unreliable in complex, detailed and critical design tasks. Some always double-checked AI outputs, and some believed that AI has limitations such as creativity, human judgment and the lack of up-to-date information.

In response to the question "Do you think AI enhance or limit your creativity?", most people expressed that AI enhances creativity, with a notable mention was "AI acts like a second brain or notebook", whereas some respondents considered AI as a limitation for their creativity due to over-reliance on AI. Some people said enhancing or limiting creativity depends on the situations and conditions. Another group of similar answers was that AI helps the work faster, allowing more time for human creativity. A few people specified that AI expands the human creativity and ideas.

4) Ethical Awareness: Findings suggested that a considerable amount of participants were aware of bias and plagiarism, and thus had an idea to cite the AI works.

Quantitative Analysis: Approximately 41% concerned that AI could be biased, while 46.2% remained neutral and only a small proportion (12.6%) reported their lack of concerns.

A large percentage (72.3%) were worried that AI-generated designs could be similar to the existing ones, potentially resulting in unintentional plagiarism. Additionally, a majority (60%) agreed that not citing the AI works is a type of plagiarism, despite the remaining (35.5%) remaining neutral.

Qualitative Analysis: First, evaluating an open-ended question "Have you ever noticed any bias in AI-generated contents?" revealed several key themes such as denial of bias by one third of respondents, followed by the occasional bias noticed by participants based on the prompts and context of use, significant number of participants believed that bias comes from training data, and ethical and copyright concerns. Second, opinions regarding the differentiation between ethical AI use and plagiarism illustrated different themes: many students considered idea generation and assistance as ethical use, while direct copying without modification was believed as plagiarism, some participants believed that ethical use and plagiarism may depend on the situation and the type of projects.

B. Controlled Experiment: Performance and Behavioral Analysis

1) Inter-Rater Reliability: Usability scores were rated independently by three evaluators across all participants and tasks, which resulted in excellent scores of reliability [27] of ICC(2,1)=0.71 and ICC(2,k)=0.88 in task 1. Similarly, task 2 score were high with ICC(2,1)=0.73 and ICC(2,k)=0.89, and task 3 was ICC(2,1)=0.72 and ICC(2,k)=0.88.

2) Finding 1: No Significant Differences in Final Design Quality: Findings displayed in Table II proved that the usage of AI does not have a significance influence on the usability quality, problem solving skills and critical thinking abilities of the participants.

Among all three design tasks, independent t-test revealed no statistically significant differences (all $p > 0.05$) in usability, problem solving and critical thinking between the AI-assisted and non-AI assisted groups, with different Cohen's d ranging from negligible to large effect size.

3) Finding 2: Potential Enhancement of Creativity in Simple Tasks: For the low complexity Task 1 (Login), AI group had a descriptively higher creativity mean score than the non-AI group. Moreover, the independent t-test showed the large effect size (Cohen's d = 1.20) along with the marginal p-value, which is approaching to significance. The similar pattern is not observed in the Task 2 and 3.

4) Finding 3: A Clear Behavioral Trade-Off: Ideation Speed vs. Iterative Refinement: During each task session of experiments, ideation time, total duration and iteration count were considered to be the criteria to determine the design iteration speed. Students in AI group tended to spend less time for their ideation than those in the Non-AI group. Specifically, AI participants often planned and brainstormed within 1-10 minutes, while non-AI participants spent 1-18 minutes. However, in terms of iteration count, AI group consistently created only one iteration per task despite faster ideation phase. Meanwhile, half of the Non-AI participants generated two iterations in the same time frame.

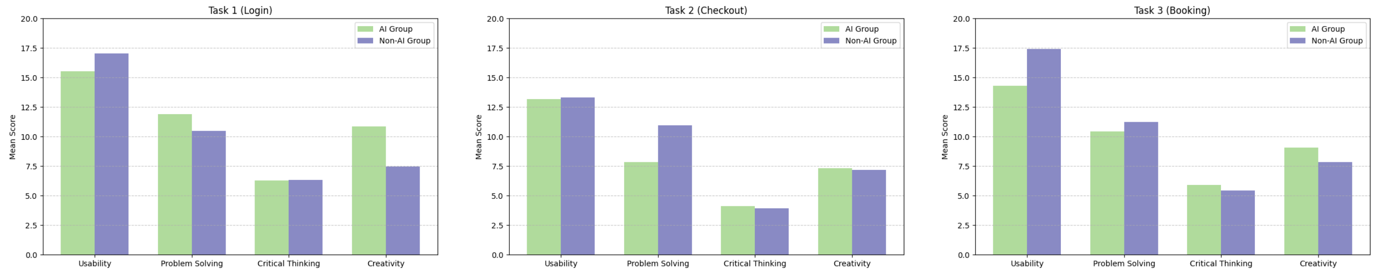
TABLE II: Results of Independent t-tests Comparing AI and Non-AI Groups Across Tasks and Dimensions

	Participant Group	Group Mean	Group SD	*t*	*p*	Cohen's *d*
Task 1 (Login)						
Usability	AI	**15.55**	**4.02**			
	Non-AI	**17.05**	**3.44**	-0.69	0.50	0.40
Problem Solving	AI	**11.89**	**2.04**			
	Non-AI	**10.50**	**2.79**	0.98	0.35	0.57
Critical Thinking	AI	**6.27**	**1.87**			
	Non-AI	**6.33**	**2.03**	-0.05	0.96	0.03
Creativity	AI	**10.84**	**1.13**			
	Non-AI	**7.45**	**3.84**	2.07	0.08	1.20
Task 2 (Checkout)						
Usability	AI	**13.16**	**4.50**			
	Non-AI	**13.33**	**3.60**	-0.07	0.94	0.04
Problem Solving	AI	**7.83**	**4.00**			
	Non-AI	**10.94**	**2.12**	-1.68	0.13	0.97
Critical Thinking	AI	**4.11**	**1.39**			
	Non-AI	**3.94**	**1.18**	0.23	0.82	0.13
Creativity	AI	**7.34**	**2.39**			
	Non-AI	**7.17**	**2.61**	0.12	0.90	0.06
Task 3 (Booking)						
Usability	AI	**14.33**	**1.80**			
	Non-AI	**17.44**	**3.82**	-1.80	0.11	1.04
Problem Solving	AI	**10.45**	**3.13**			
	Non-AI	**11.22**	**1.38**	-0.55	0.59	0.32
Critical Thinking	AI	**5.89**	**2.28**			
	Non-AI	**5.45**	**1.19**	0.42	0.69	0.24
Creativity	AI	**9.05**	**3.21**			
	Non-AI	**7.83**	**2.46**	0.74	0.48	0.73

C. Post-experiment Survey

Quantitative Analysis: Majority of participants (75%) who completed the onsite experiment agreed or strongly agreed that AI enhances the UX learning experience. However, 16.7% were neutral about the statement and only one participant (8.3%) disagreed with it.

When asked about the participants' comfort level of using AI-generated content in the commercial projects, the majority



(a) Task 1: Login (b) Task 2: Shopping Checkout (c) Task 3: Transportation Booking

Fig. 2: Bar Charts for the Descriptive Mean Scores (AI vs. Non-AI) Across Four Dimensions for Each Task

(58.3%) responded that it is context-dependent, compared to 25% who were comfortable and 16.7% who felt the opposite. The participants are equally divided on “Maybe” and “Yes” to the question on whether they will use AI in their future UX/UI projects.

Qualitative Analysis: Participants were continued to express their reasons to choose ”Yes” or ”Maybe” to the future use of AI. Most answered they would like to use AI for idea generation, brainstorming, checking mistakes humans overlooked.

Participants emphasized the plagiarism and their concern about stealing or copying other existing design ideas, upon asking “What ethical concern should be addressed before integrating AI into UX education?”.

1) Perspectives of AI-assisted Group on Bias and Plagiarism: AI-assisted participants acknowledged that AI version was similar to existing real-life applications, which raised the plagiarism and bias issues. Additionally, they mentioned the lack of emotional touch in the AI designs.

Quantitative Analysis: According to the bias and plagiarism survey there was a balanced split between the participants who believe both design from the AI generated version and version the participants have refined, which was modeled after AI generated results, were inclusive. A high number (66.7%) of participants also agreed that there are similarities between real-life applications and AI generated versions.

*Qualitative Analysis:*The participants also expressed that the AI generates designs that are generalized and stereotypical. The participants also mentioned that AI has no consideration for those who have visual impairments since it is most often generated only in basic colors. In order to reduce biases in the AI generated personas, participants suggested using personal judgment and working in comparison with existing software. Moreover, becoming limited in critical design thinking, facing plagiarism and showing lack of emotional touch in design are some of the risks the participants think about directly copying AI generated results without modifications. To maintain originality of the design, the participants encouraged adding personal touches such as font styles and colors to fit the theme of the application system.

V. Discussion

The results of quantitative and qualitative analyses of mix-method study offers a nuanced understanding of how generative AI tools have an influence on the UX design education in terms of usability, problem solving, critical thinking, creativity and ethical awareness.

A. Impact on Performance and Design Efficiency

The study confirms that AI helped in the design ideation faster by allowing the participants to complete the ideation and brainstorming faster in during experiment sessions, allowing students to start developing the prototype faster. This finding validates the previous studies [11]–[13], which highlighted in their studies about integrating AI tools improved the ideation by decreasing the design development duration.

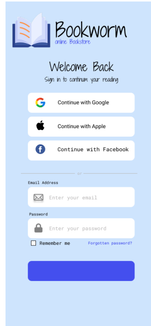
However, notable finding from the controlled experiment highlight the trade-off. This study also found that faster ideation phase does not equate to the better quality and usability of designs. Conversely, non-AI participants completed more iterations and refinements than their AI-assisted counterparts. Non-AI participants were more likely to revisit and refine their own designs. This emphasizes a crucial educational challenge, in which AI might assist primarily in the faster initial speed, rather than prioritizing the actual reflective practice of iterative development and refinement that is essential for profound design knowledge and outcomes.

B. Impact on Critical Thinking and Creativity

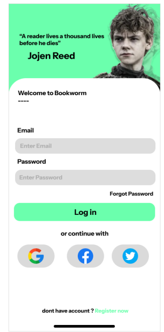
This study help showed that AI assistance creates a potential enhancement in the creativity, but only in easier and simpler UI task with low complexity. The quantitative results showed that AI group has a descriptively higher creativity than non-AI group, with larger effect size. Qualitative responses confirmed this, with participants expressing AI as “second brain” which enhances creativity. This implies UX educators should teach students that AI tools as a “brainstorming partner” rather than a solution generators.

C. Ethical Awareness

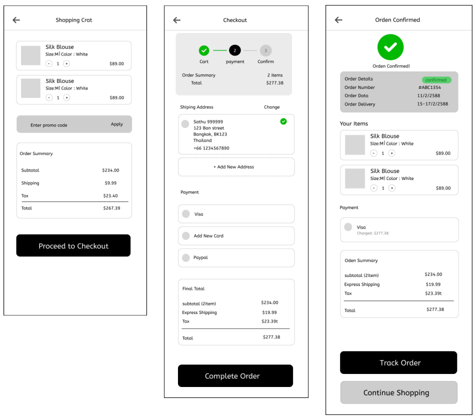
Participants demonstrated the high awareness of ethical issues such as plagiarism and bias, with the substantial proportion of respondents stated the need to double-check AI generated contents. Experiment group participants also mentioned the lack of human emotional touch in AI designs, and showed their concerns towards copying AI contents without citing or modification. While the high level of ethical awareness found in most respondents were encouraging, it also highlights the conflict between between efficiency of AI usage and the risk



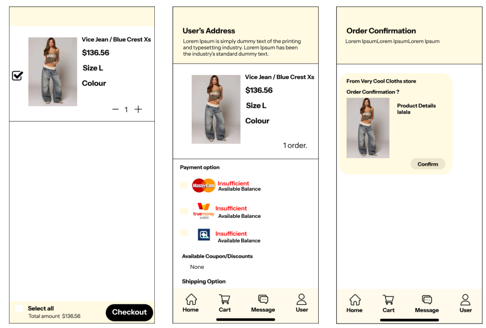
(a) AI-assisted: Login



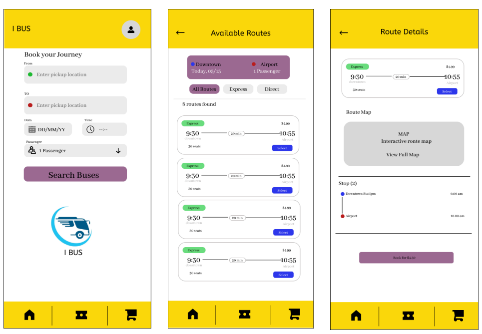
(b) Non-AI-assisted: Login



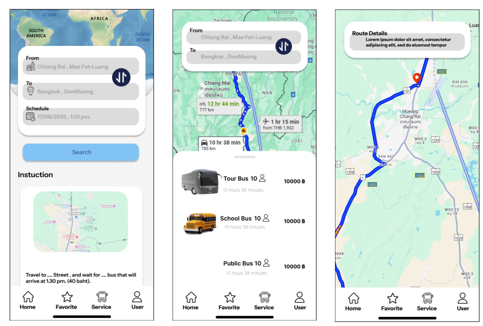
(c) AI-assisted: Shopping Checkout Flow



(d) Non-AI-assisted: Shopping Checkout Flow



(e) AI-assisted: Transportation Booking App



(f) Non-AI-assisted: Transportation Booking App

Fig. 3: Comparison of Sample Figma Designs: Row 1 shows login screens; Row 2 shows checkout, shipping, and confirmation screens; Row 3 shows search and route details. Each row contrasts AI-assisted (left side) and Non-AI-assisted (right side) designs.

of academic misconduct. These findings align with the broader criticism of Shneiderman and Lu in their previous studies [18], [19]. Therefore, educators are advised to implement the literacy on the digital ethics, responsible AI use and proper citation in the UX education. This includes instructing students to critically analyze AI outputs, cite AI generated contents transparently and balance their reliance on AI with their creativity skills.

D. Implication for UX Education

The study suggests that integrating AI into the UX/UI design education could enhance the efficiency and ideation with the simultaneous positive potential effect on creativity and ethical considerations. Thus, UX educators should encourage students on generating their own personal insights combined with AI-generated suggestions, to mitigate the unintentional plagiarism, bias and emotional touch.

E. Limitations and Future Research

The primary limitation of this study is the small sample size of students, drawn exclusively from a single university, which may limit and restrict the generalization and the comprehensive understanding on the overall population of UX students. Additionally, this study excluded the System Usability Scale (SUS) analysis and usability testing with the potential end users, by analyzing the pain points of each design developed by the participants.

Future research should be conducted on multiple diverse groups of students from different institutions or even cross-national contexts. As this paper utilized ChatGPT and UX Pilot as AI assistant tools, it is highly recommended to compare the efficiency and effectiveness of the other different AI tools as well. Usability testing with the potential users would be suggested to include in the future researches. Moreover, ablation studies that compare the effects of different AI tools would be helpful in identifying further effectiveness of specific AI tools in UX education. Beyond the tasks in the methodology, investigating AI effect on the collaborative work flow and real-life projects would provide further insights into the impact of AI on UX education and design iterations.

VI. Conclusion

This study explore a new perspective on the impact of AI in UX Education, on the overall learning process, creativity, design iteration speed and problem solving skills, combining both qualitative and quantitative analysis to detailed understanding. The conclusion can be drawn from this study that while AI helps in UX education for their efficiency in iteration speed and as a supportive tool for ideation, its impact on the critical design outcomes, usability, critical and problem solving proves to be very minimal. As different kinds of AI tools become integrated into the UX education, the critical role which UX/UI teaching should be emphasize is the cultivation of the critical judgment and human-center principles in design learning. This study suggests that recent AI tools serve better as assistant in design ideation and iteration rather than standard tools for design principles education.

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